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Paleomagnetism, anisotropy of magnetic susceptibility, and argon–argon geochronology of the Clearwater anorthosite, Saskatchewan, Canada

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Abstract

The Clearwater anorthosite in Saskatchewan is a major gabbro–anorthosite layered intrusion lying within the Archean Rae Province along the northwestern hinterland of the Paleoproterozoic Trans-Hudson Orogen (THO). The intrusion, which could be at least 2.4 Ga old, carries a well-defined remanent magnetization with a direction of Declination = 112° and Inclination = +38° ($k = 316$, $\alpha_{95} = 3.1°$, $N = 8$). The corresponding paleomagnetic pole (Plat. = 6.5°N, Plong. = 48.2°W, dp = 2.2°, dm = 3.7°) is the first one obtained from the Archean Western Granulite Domain of the Rae Province. Field observations, supplemented by anisotropy of magnetic susceptibility data, show that the magnetization post-dates ductile deformation of the anorthosite. Argon–argon plateau dates of 1917 ± 7 Ma for hornblende and 1839 ± 5 Ma for biotite are interpreted as cooling ages, with the hornblende age approximating the time of remanence. The pole position, and others of comparable age from the northwestern margin of the THO form a loose cluster, the mean of which is about 90° of arc away from that of a broadly contemporaneous group of poles from the Superior Province. This relation supports previous interpretations of paleomagnetic data that, prior to the THO at about 1.9 Ga ago, a major ocean (the Manikewan ocean), at least 5000 km across, separated the Superior and Rae/Hearne Provinces. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Ar–Ar dating; anisotropy of magnetic susceptibility; paleoproterozoic anorthosite

1. Introduction

The Clearwater anorthosite complex (Scott, 1985) is a N–S to NE–SW trending intrusion about 24 km long and 4 km wide that occurs in the Western Granulite Domain (WGD) of the Rae Province in northern Saskatchewan (Figs. 1 and 2). The granulite terrane has been interpreted as an essentially cratonic Archean hinterland bordering the western side of the Trans-Hudson Orogen (THO) (Lewry and Sibbald, 1980). The anorthosite lies about 20 km west of the Virgin River shear zone, a major tectonic break which represents part of the Snowbird Line, a 1500 km long zone of granulitic mylonites and associated anorthositic rocks that forms the boundary between the Archean Rae and Hearne provinces (Hoffman, 1990). U–Pb studies suggest that the WGD suffered a middle to deep crustal granulite metamor-

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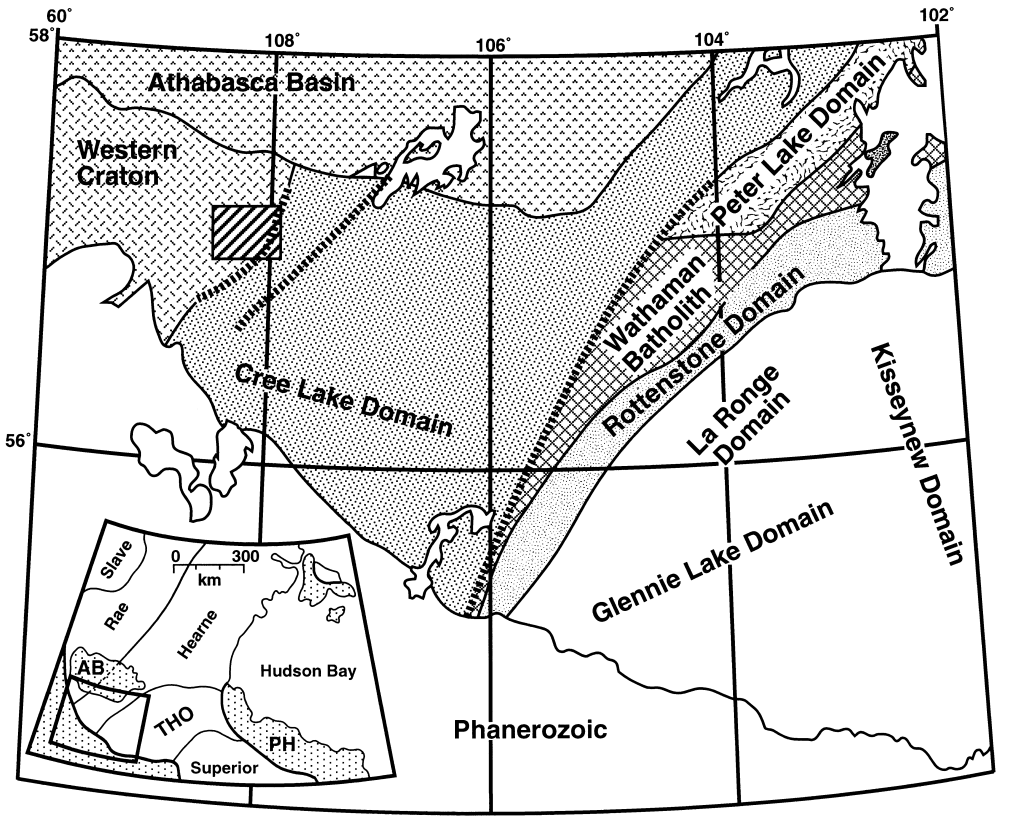


Fig. 1. Map showing the location of the study area with respect to the Trans-Hudson Orogen. Hatched lines are faults. The inset diagram shows the location of the map in the larger context of the geology of central Canada. Also shown are the locations of the Slave, Rae, Hearne and Superior provinces, the Trans-Hudson Orogen (THO), and the cover of unmetamorphosed sedimentary strata: *PH* = Phanerozoic; *AB* = Mesoproterozoic rocks of the Athabasca Basin.

phism at about 2.0–2.3 Ga, but that the high-grade metamorphic overprint occurred at shallower crustal levels in the Hearne Province, suggesting that the Rae Province was a Tibetan-like uplift, thrust eastwards along the Virgin River fault (Fig. 2) and over the Hearne at about 2.0–2.3 Ga (Bickford et al., 1994).

Stauffer (1984) proposed the existence of a major ocean (the Manikewan) between the Superior and Rae/Hearne provinces prior to the THO. On the basis of U–Pb dating of rift-related mafic rocks on opposite sides of the THO, the ocean opened at about 2.1 Ga (Halls and Heaman, 1997) and closed at about 1.8 Ga corresponding to the collisional phase of the orogen. Subsequent paleomagnetic work on rocks in the age range of 1850–1900 Ma, by Symons and co-workers (e.g., Dunsmore and Symons, 1990; Symons, 1991, 1994; Symons et al., 1994, 1995), has shown that the collisional suture lies along the eastern boundary of the Wathaman batholith (Fig. 1), whereby poles from units in the Wathaman batholith and Peter Lake domain (Fig. 1) are typical for the Rae/Hearne provinces whereas those to the east, including several within the THO, appear to have been part of the Superior Province. The suture has been therefore regarded as the closure scar of the Manikewan ocean (e.g., Symons et al., 1994) and the Wathaman batholith as the product of Andean-type subduction along the western margin of the ocean (Meyer et al., 1992). According to Symons et al. (1995) the paleomagnetic data are consistent with the Manikewan ocean having a maximum width of about 5000 km.

Reliable paleomagnetic data for the period

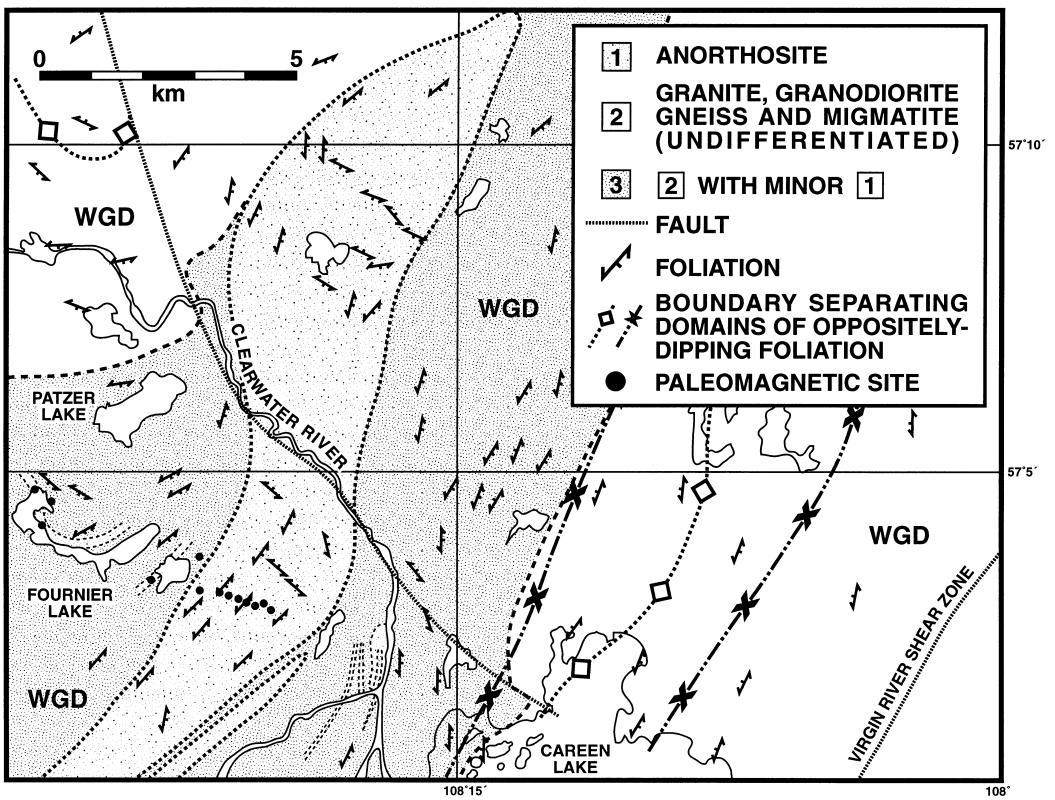


Fig. 2. Geological map showing foliation patterns within the Clearwater anorthosite and the enclosing gneisses of the Western Granulite Domain (*WGD*). The positions of the Virgin River shear zone and paleomagnetic sampling sites are also shown.

prior to the opening of the Manikewan ocean are extremely few, particularly in the Rae/Hearne provinces. The Clearwater anorthosite offers the possibility of a pre-2.1 Ga age because it appears interlayered and folded with Archean gneisses of the Western Granulite Domain and may have been uplifted before 2.1 Ga. If the Superior and Rae/Hearne provinces contain magmatic products related to the opening of the Manikewan ocean, then prior to 2.1 Ga they may have been joined. In this paper we test this idea through the use of paleomagnetism, anisotropy of magnetic susceptibility (AMS) and Ar–Ar geochronology, by searching for a possible pre-2.1 Ga uplift-related remanence that might be compared to well-dated magnetizations in the Superior Province such as those from the 2167 Ma Biscotasing and 2216 Ma Seneterre dyke swarms (Buchan et al., 1993).

2. Geology

The Clearwater complex comprises a layered cumulate sequence dominated by anorthosite and anorthositic norite to gabbro (Scott, 1985; Hulbert, 1988). The rocks are often extremely fresh with primary calcic plagioclase, clinopyroxene, orthopyroxene and olivine. The complex exhibits primary layering, which dips on average about 35°SE and is normally expressed by thin centimetre-wide bands relatively enriched in ferromagnesian minerals (up to about 10%). The body contains clear evidence of multiple intrusion and becomes more mafic and coarser-grained towards the east (Crocker et al., 1993). The exact dimensions of the complex are unknown because contacts are poorly defined. West of the intrusion, around Fournier Lake, anorthositic bands indistinguishable from the main body occur

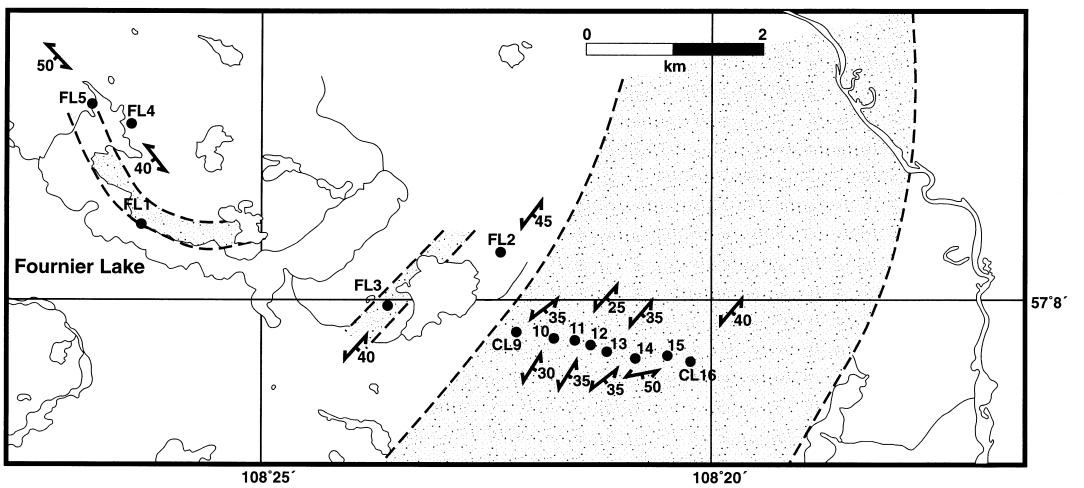


Fig. 3. Map showing the location of paleomagnetic sampling sites and the mean foliation orientation at each site. Anorthosite units are stippled.

interleaved with garnet and amphibole-bearing granitic gneisses, the foliation of which remains parallel to the layering in the anorthosite, despite regional changes in strike (Figs. 2 and 3). Along the eastern margin, anorthositic breccias are reported (Hulbert, 1988) and the anorthosite is again concordantly interlayered with more felsic units including garnetiferous tonalite and granite (Crocker et al., 1993). In view of its intimate structural relationship with the gneisses, the granulite grade of its host rocks and its possible upper mantle origin (R. Emslie in Scott, 1985), the anorthosite could be part of the Western Granulite Domain and hence Archean in age. However, the absolute age of the intrusion is uncertain; two Sm–Nd ages give different results: 1846 ± 83 Ma (Crocker et al., 1993) and 2408 ± 30 Ma [1] (Hulbert, 1988). NW–SE-trending shear zones are present in the anorthosite and are associated with drag folding of the layers which may locally rotate their strike towards a NW direction. In the Western Granulite Domain retrograde metamorphism to amphibolite facies increases eastwards from the Clearwater anorthosite towards the Virgin River shear zone, the last movement of which is after 1.82 Ga and corresponds to a subduction or later collisional phase of the THO (see Crocker et al., 1993, for summary). Small pegmatite dykes and granitic sheets locally cut the anorthosite and may be related to this event (Crocker et al., 1993).

3. Methodology

3.1. Paleomagnetism

About 120 samples, each oriented using both magnetic and sun compasses, were field-drilled at 13 sites mainly in anorthosite, along a 7.5 km long traverse oriented normal to strike (Fig. 3). Since outcrops often form high knolls, care was taken to sample around their base to avoid ubiquitous lightning strikes at higher elevations which caused spalling of the weathered surface and differences between magnetic and sun compass readings. The samples were subject to alternating field (AF) and thermal demagnetization using Schonstedt demagnetizer models GSD-1 and TSD-1, respectively. Remanent magnetization was measured on a modified Digico magnetometer using the method outlined by Halls (1986). Remanent magnetization directions were analysed using the remagnetization circle analysis of Bailey and Halls (1984) and Principal Component Analysis (Kirschvink, 1980) with a modified least-squares fitting procedure (Halls, 1986). Means were calculated using the statistics of Fisher (1953).

[1] Due to a mis-print, the age appears incorrectly as 2480 Ma in the original report.

3.2. Anisotropy of magnetic susceptibility (AMS)

AMS was measured for all samples by a Sapphire Instruments susceptibility bridge model SI-2B, using seven sample orientations as originally proposed by Nye (1957) and advocated for routine work by Borradaile and Stupavsky (1995), with subsequent data reduction combining the techniques of Ernst and Pearce (1989) and Jelinek (1978).

3.3. Argon–argon geochronology

Six Ar–Ar analyses (four step-heating, and two total fusion) were carried out on mineralogically fresh biotite and hornblende separates obtained from three sites in the Clearwater anorthosite (CL15, CL11 and FL3; Fig. 3). The mineral separates were produced by crushing, sieving and final hand-picking using a binocular microscope. The separates (ca. 20 grains) were wrapped in aluminum foil packets and placed into an aluminum canister, along with aluminum foil packets containing the monitor hb3gr (1072 Ma). The positions of all packets were carefully recorded to correct for spatial variations in fast neutron flux inside the reactor. The aluminum can was irradiated for 200 h in site 5C of the McMaster University research reactor. Following irradiation, samples were loaded into holes drilled in a copper sample holder, with each hole holding from one to fifteen mineral grains. Samples were then either melted (total fusion) or heated at sequentially higher temperatures (step-heating) using an 8-W Lexel 3500 argon laser on-line to a MAP 216 mass spectrometer. Mass spectrometric ratios were corrected to an $^{40}Ar/^{36}Ar$ atmospheric ratio of 295.5, and were corrected for neutron-induced interferences and system blanks.

4. Results

4.1. Petrography

All samples from the Clearwater anorthosite contain more than 90% plagioclase. Ferromagnesian minerals (mainly pyroxene and a green amphibole with lesser biotite and relict olivine) constitute less than 10%, and accessory magnetite less than about 2%. The anorthosite is generally fresh with pyroxene locally altered to amphibole and feldspar with no more than about 10% saussuritization. On scales of several tens of metres the pyroxene can change appearance from pervasive alteration to amphibole, to complete freshness. The amphibole, where it rims the pyroxene, or as apparently discrete grains, is fresh and pleochroic in shades of green. Where pyroxene is partially altered, fresh and altered parts alternate to give a fibrous appearance. The degree of alteration remains constant along the sampling traverse; therefore, the increase in retrograde metamorphism reported by Crocker et al. (1993) towards the Virgin River shear zone occurs farther east. A search was made of biotite and amphibole grains for pleochroic haloes that might indicate zircon or other minerals suitable for U–Pb dating, but none were found.

4.2. Paleomagnetism

An extremely hard characteristic remanent magnetization **A** is preserved in the anorthosite. The remanence is particularly stable and retains a uniform direction of magnetization, revealed either by AF or thermal demagnetization (Fig. 4). Typically the NRM intensity (J_0) averages about 3 A/m and J_{40}/J_0 is about 0.4, where J_{40} is the magnetization intensity left after a demagnetizing at 40 mT. The thermal demagnetization data indicate that the carrier of the magnetization is magnetite and the high coercivity of the remanence suggests the magnetite to be PSD and/or SD in grain size. The magnetite occurs in the anorthosite as relatively rare interstitial grains but also probably as a fine dusting within the plagioclase which appears slightly cloudy.

After removal of low coercivity (less than 20 mT) or lower unblocking temperature (less than 400ºC) components, most samples exhibit **A** and therefore the analyses of the paleomagnetic data are straightforward. Within the main body of the anorthosite **A** has an extremely consistent direction which is independent of the orientation of the foliation (Fig. 5). In particular, at site CL15 where NW–SE-trending shear zones have caused drag folding of the anorthosite, the direction remains invariant demonstrating that the remanence was acquired after this deformation.

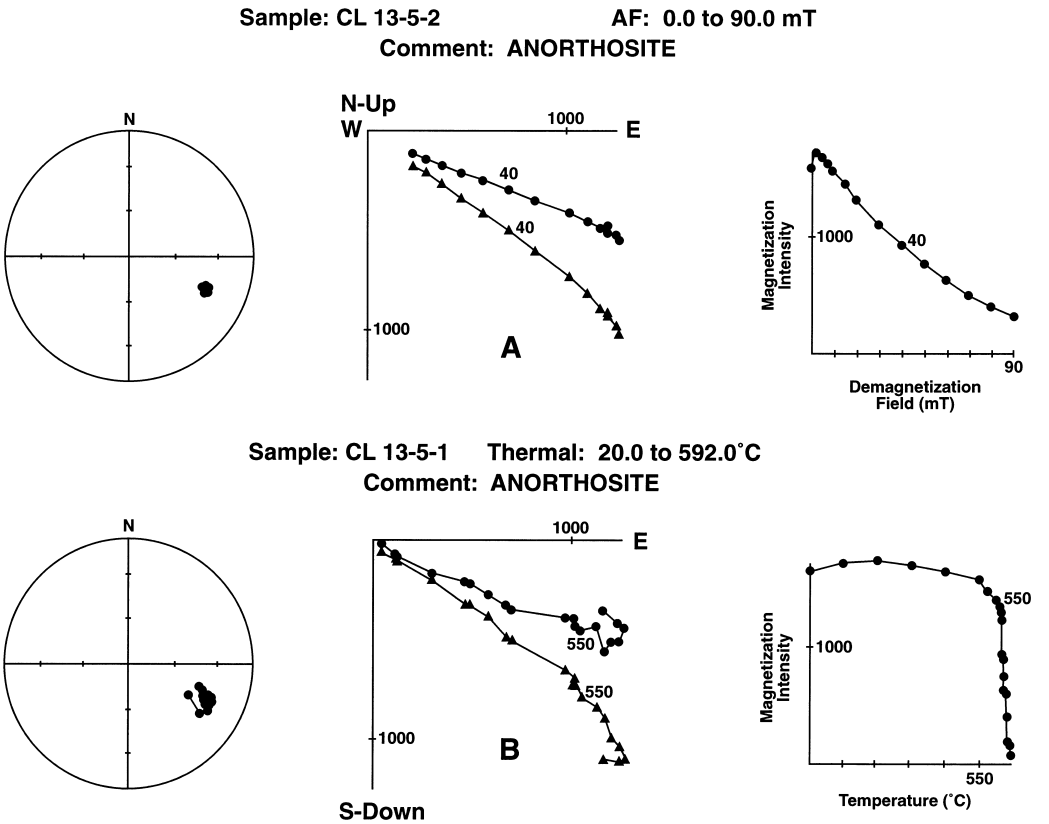


Fig. 4. Examples of equal-area stereonets, vector diagrams and magnetization intensity decay plots of (A) alternating field and (B) thermal demagnetization, showing the typical stable behaviour of the characteristic remanence. For equal-area stereonets, solid symbols indicate a positive (downward-pointing) magnetization; for vector diagrams (which portray changes in magnetization intensity and direction of the natural remanence after each demagnetization step, projected on two orthogonal planes), dots are plotted on a horizontal projection and triangles on a vertical projection. In vector diagrams and decay plots, the magnetization intensity is in units of A/m $\times$ 10^{-3}.

The paleomagnetic results (Table 1) show that the least-dispersed data come from sites within the main anorthosite body. Two sites (FL2 and 5) which are from mixed anorthosite–gneissic rocks are less stable with more erratic data, but give a mean direction similar to the others. Two other sites (FL1 and 4) also gave the same direction but this was confined to one or two anorthositic samples; remaining samples in gneiss were weak and unstable. For the purpose of averaging, only sites with data from seven or more independently oriented samples were used. Sites 15 and 16 were combined because of their close proximity.

Therefore for a total of eight site directions (Fig. 6), the mean obtained was Dec. = 112º, Inc. = 38º, k = 316 and α_{95} = 3.1º, which represents the average remanence direction along a traverse across the anorthosite of about 3 km. The direction yields a paleomagnetic pole at Plat. = 6.5ºN, Plong. = 48.2ºW, dp = 2.2º, dm = 3.7º.

4.3. Anisotropy of magnetic susceptibility

Bulk susceptibility values (Table 2) are consistent with the AMS signature being carried by magnetite which is also visible in thin section. The AMS results (Table 2; Fig. 7) show that the susceptibility ellipsoid is oblate with an extremely well-defined minimum susceptibility axis normal to foliation. A weaker lineation is also present with the maximum susceptibility axis approximately parallel to the regional strike of the anorthosite (cf. Figs. 2 and 7).

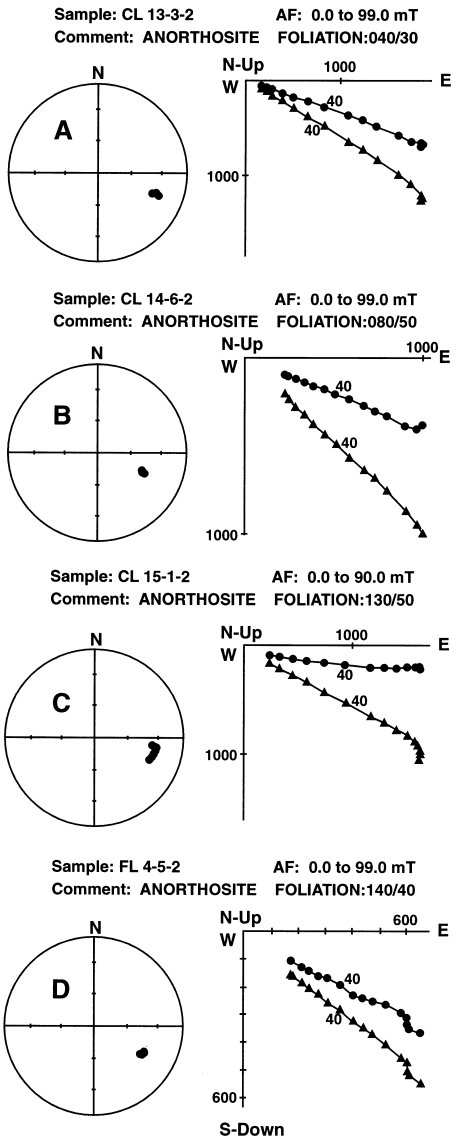


Fig. 5. Equal-area stereonets and vector diagrams showing that the direction of remanence is independent of the orientation of foliation (given at the top of each diagram in the form S/D where S is strike and D is dip to the south or southeast). Symbols are as in Fig. 4.

This may reflect the original flow of the magma, but it is also the direction expected from a re-alignment of feldspar laths under east to east-southeast compression during overthrusting along the Virgin River shear zone. Lineation predominates only at site 15 (Fig. 8) where it parallels the strike of small SE–NW-trending ductile faults. Although the lineation is not visible in the field, it is caused presumably by sub-horizontal shearing along the faults during compression.

Table 1
Paleomagnetic results

Site	F (º)	N_S	N_R	N	Dec. (º)	Inc. (º)	k	α_{95} (º)
FL1	–	4		3	90.8	45.5	56	16.6
FL2	40/45	9		4	129.6	52.8	223	6.2
FL3	40/40	12	5	7	119.2	34.7	62	6.5
FL4	140/40	6		1	121.5	37.0	–	–
FL5	140/50	15		6	129.9	35.4	64	8.4
CL9	40/30	24		23	109.6	36.8	164	2.4
CL10	35/30	29	6	23	108.1	35.9	92	3.0
CL11	35/35	13		7	109.1	38.5	86	6.5
CL12	35/35	14	4	10	114.9	42.8	165	3.4
CL13	40/35	8		7	109.9	33.6	136	5.2
CL14	50/30 [a]	17		14	116.4	39.9	75	4.6
CL16 [b]	40/40	11		9	111.1	41.3	61	6.7

Symbols: F = strike/dip of foliation; N_S = number of independently oriented samples; N_R and N = number of independent samples giving remagnetization circles and stable end-points, respectively; Dec. = mean declination; Inc. = mean inclination; k = precision parameter of Fisher (1953); α_{95} = radius in degrees of 95% confidence cone about the mean.

[a] Samples were also collected where the foliation was 80º/60º.

[b] The mean includes three samples from site CL15 which has a mean foliation direction of about 120º/50º.

4.4. Argon–argon geochronology

Results from the Ar–Ar analyses are presented in Table 3, and the step-heating age spectra are shown in Fig. 9. In the age spectra, the vertical heights of the bars for each fraction represent $\pm 2\sigma$ SE, suitable for comparing within-spectrum dates, where uncertainty in the measurement of flux is not a factor (i.e., error in $J = 0$). A conservative 0.5% error in J estimates can be used for between-spectra age comparisons of samples irradiated together; this corresponds to an additional ca. ± 5 Ma (2σ) error in step dates.

4.4.1. Hornblende

The hornblende from sample CL11-11 (site CL11) in the Clearwater anorthosite yielded an excellent plateau date of 1917 ± 07 Ma (Table 3 and Appendix A) for the last four steps of the age spec-

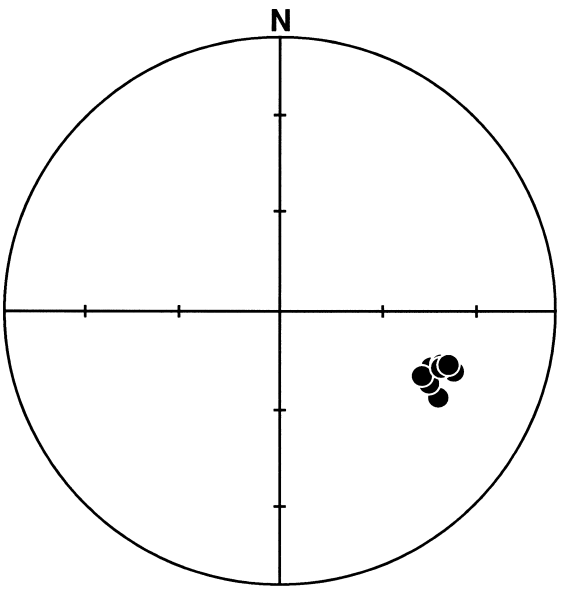


Fig. 6. Equal-area plot of the magnetization directions from the eight sites used to estimate the mean for the Clearwater anorthosite.

trum (62% of the ^{39}Ar released; Fig. 9a). These four steps have ^{37}Ar/^{39}Ar ratios that overlap within error, and that correspond to a Ca/K ratio of ~6–7 (see Appendix A). The high date in the first step is attributed to excess argon.

Table 3
Results of argon–argon analyses

Sample	N_g	N_s	T_i	T_f	f (^{39}Ar)	T_p
Hornblende						
CL11-11	15	7	2035 ± 7	–	63	1917 ± 7
Biotite						
CL15-3	10	7	1829 ± 5	–	95	1840 ± 5
FL3-11	6	7	1833 ± 7	94	1838 ± 7	
FL3-3-2	4	2	1795 ± 12	–	95	1805 ± 12
CL15-3	1	1	–	1810 ± 6	–	–
FL3-11	1	1	–	1823 ± 5	–	–

T_i = integrated date in Ma; N_g = number of grains fused; N_s = number of heating steps; T_f = total fusion date in Ma; f (^{39}Ar) = percent ^{39}Ar released on the plateau; T_p = plateau date in Ma.

4.4.2. Biotite

The eight-step age spectra for biotite samples CL15-3 (site CL15) and FL3-11 (site FL3) are virtually identical (Fig. 9b,c). With the exception of the young dates for the first 5% of ^{39}Ar released, the remaining steps yield plateau dates of 1840 ± 05 Ma (CL15-3) and 1838 ± 07 Ma (FL3-11). Total fusion dates on single biotite grains from these two samples are close to this plateau date (1823 ± 05 Ma and 1810 ± 06 Ma, respectively; Table 3). A less detailed, three-step, analysis of biotite FL3-3-2 at site FL3 gave a two-step plateau date of 1804 ± 25 Ma

Table 2
Anisotropy of magnetic susceptibility data

Site	N	K_{max}	Dec.	Inc.	$A_{95_{max}}$	$A_{95_{min}}$	K_{min}	Dec.	Inc.	$A_{95_{max}}$	$A_{95_{min}}$	K_{int}	K_{mean}
FL1	4	1.099	131	59	79	27	0.879	324	31	58	36	1.021	0.040
FL2	8	1.057	192	35	43	12	0.938	310	34	30	11	1.005	0.028
FL3	13	1.042	188	40	30	8	0.958	329	42	17	10	0.999	0.006
FL4	6	1.041	188	47	52	11	0.962	50	34	43	11	0.997	0.002
FL5	14	1.078	182	39	14	8	0.907	16	50	21	6	1.014	0.007
CL9	24	1.079	184	14	11	4	0.913	308	65	8	4	1.008	0.018
CL10	27	1.049	178	20	19	7	0.934	303	58	8	6	1.017	0.019
CL11	13	1.053	131	29	25	7	0.923	324	61	8	5	1.024	0.006
CL12	14	1.068	178	23	24	8	0.899	326	64	8	5	1.033	0.009
CL13	8	1.074	207	12	40	8	0.877	317	59	8	7	1.049	0.009
CL14	16	1.075	135	35	19	6	0.884	334	54	6	3	1.042	0.010
CL15	5	1.063	116	23	36	24	0.959	262	63	75	21	0.978	0.005
CL16	6	1.067	178	27	48	9	0.898	318	57	12	10	1.036	0.012

N = number of independently oriented samples; K_{max} = axial length of maximum susceptibility ellipsoid, where axes are unity for isotropism; K_{mean} = mean susceptibility in SI units; Dec., Inc. = declination, inclination (positive); $A_{95_{max,min}}$ = radii of 95% confidence ellipse about the mean direction of susceptibility axis.

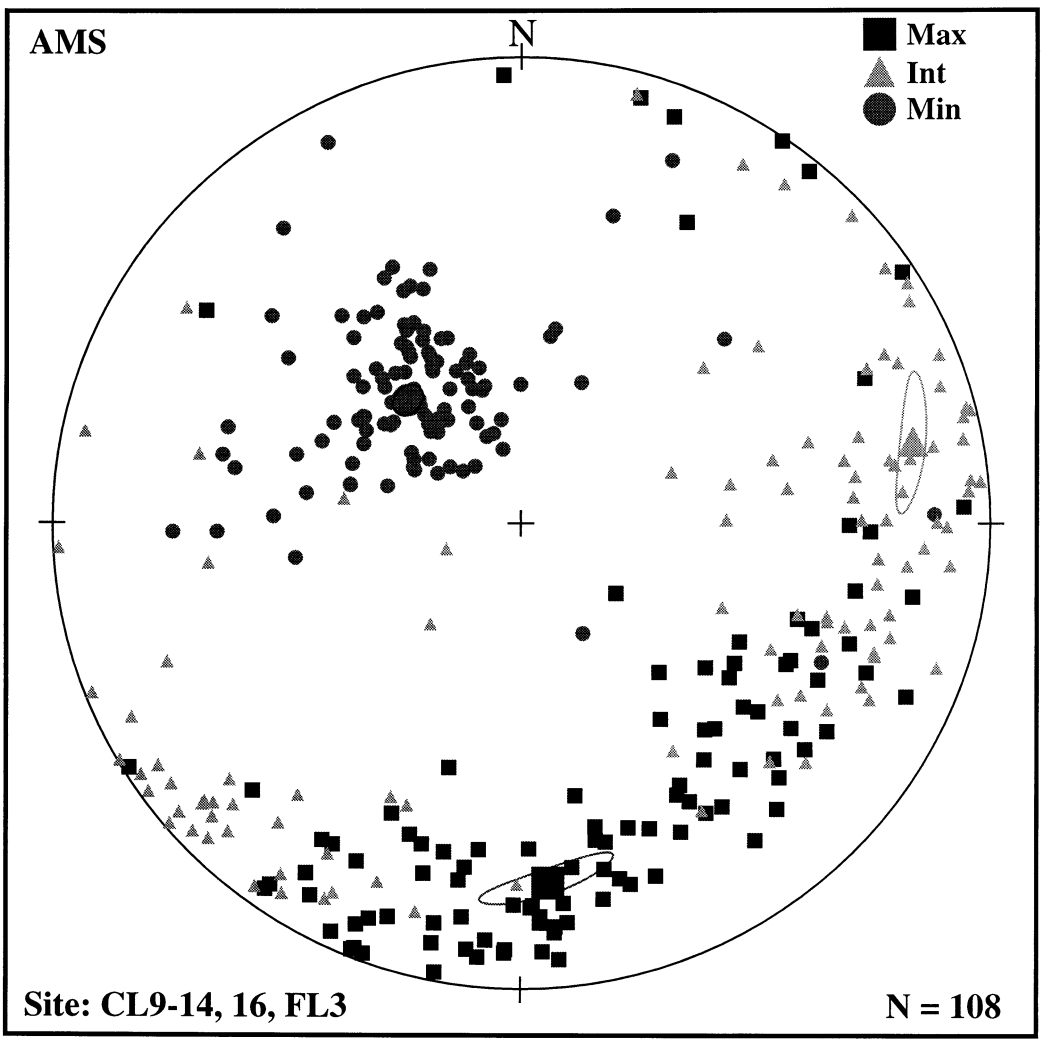


Fig. 7. AMS results for all samples from sites used to obtain the paleomagnetic pole. Ovals are 95% confidence limits about each axis of the susceptibility ellipsoid. Data are plotted on the lower hemisphere of the stereonet.

(Fig. 9d), which statistically overlaps at 2F with the plateau date of FL3-11.

5. Discussion and conclusions

The Ar–Ar dates provide constraints on the cooling history of the Clearwater anorthosite. The plateau date of 1917 ± 07 Ma (Fig. 9a) for hornblende from site CL11 is interpreted as the time of last closure of the hornblende to argon loss (~500ºC). Furthermore, the excellent 1840 ± 05 and 1838 ± 07 Ma plateau dates (Fig. 9b,c) for biotite from sites CL15 (sample CL15-3) and FL3 (sample FL3-11) are interpreted as the best estimate of the last closure of biotite (~280–300ºC).

The 1.92 Ga hornblende and 1.84 Ga biotite closure ages may have been imprinted during slow cooling of the Clearwater anorthosite following its emplacement and the high-grade metamorphic event caused by eastward thrusting along the Virgin River fault of the Rae Province over the Hearne at about 2.0–2.3 Ga (Bickford et al., 1994). This would correspond to an average cooling rate (between 500ºC and

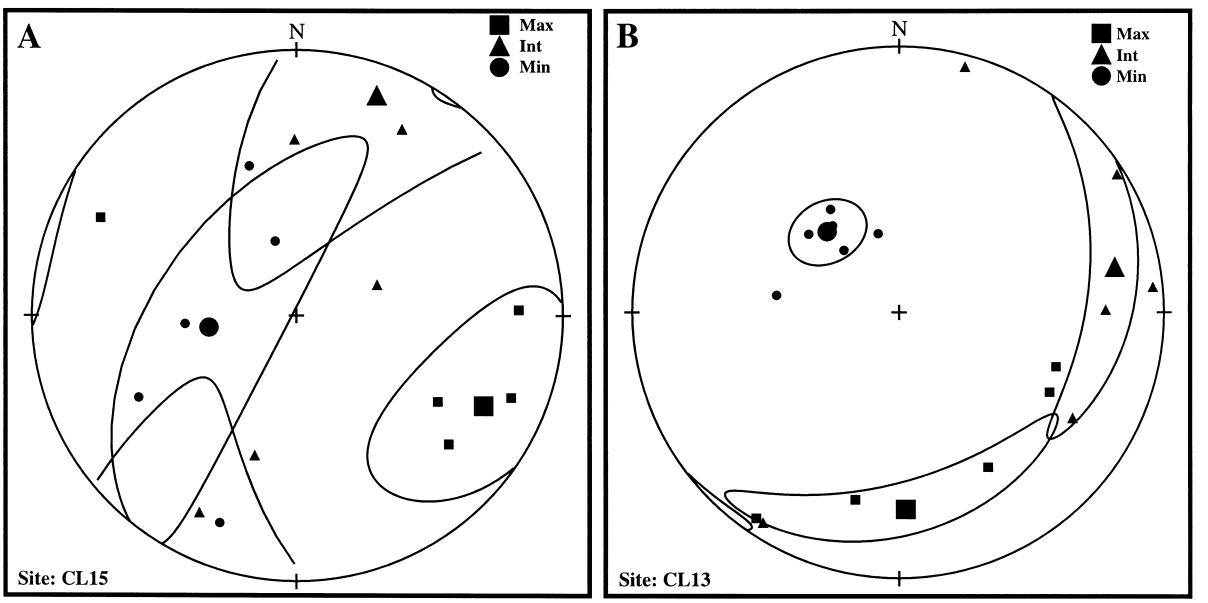


Fig. 8. AMS data showing the pronounced SE lineation at site CL15 (A) compared to the strong foliation at site CL13 (B) which is typical of all other sites. Ovals are 95% confidence limits about each axis of the susceptibility ellipsoid. Data are plotted on the lower hemisphere of the stereonet.

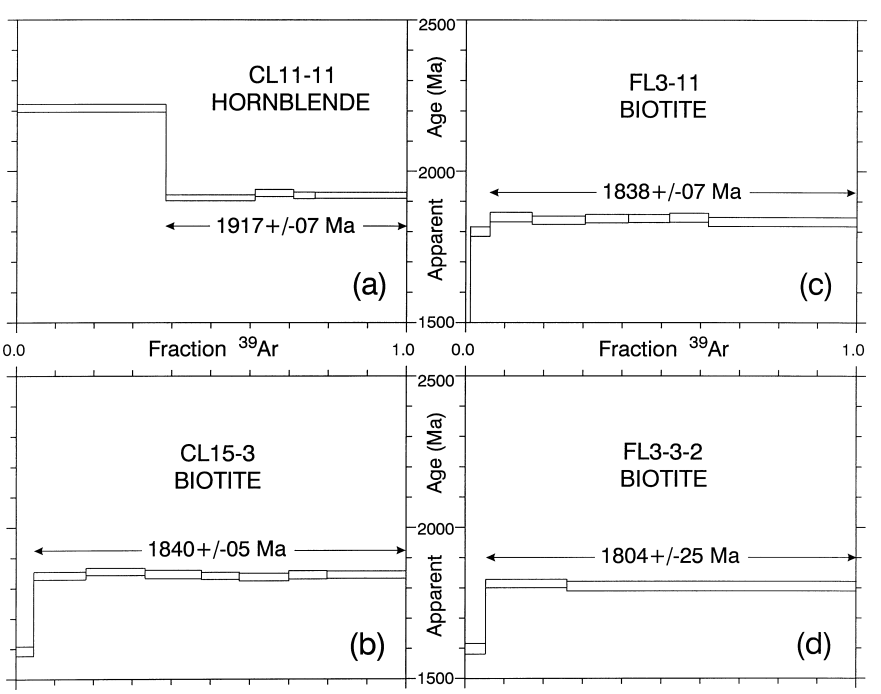


Fig. 9. Ar–Ar age spectra for (a) hornblende CL11-11, and for biotites (b) CL15-3, (c) FL3-11, and (d) FL3-3-2.

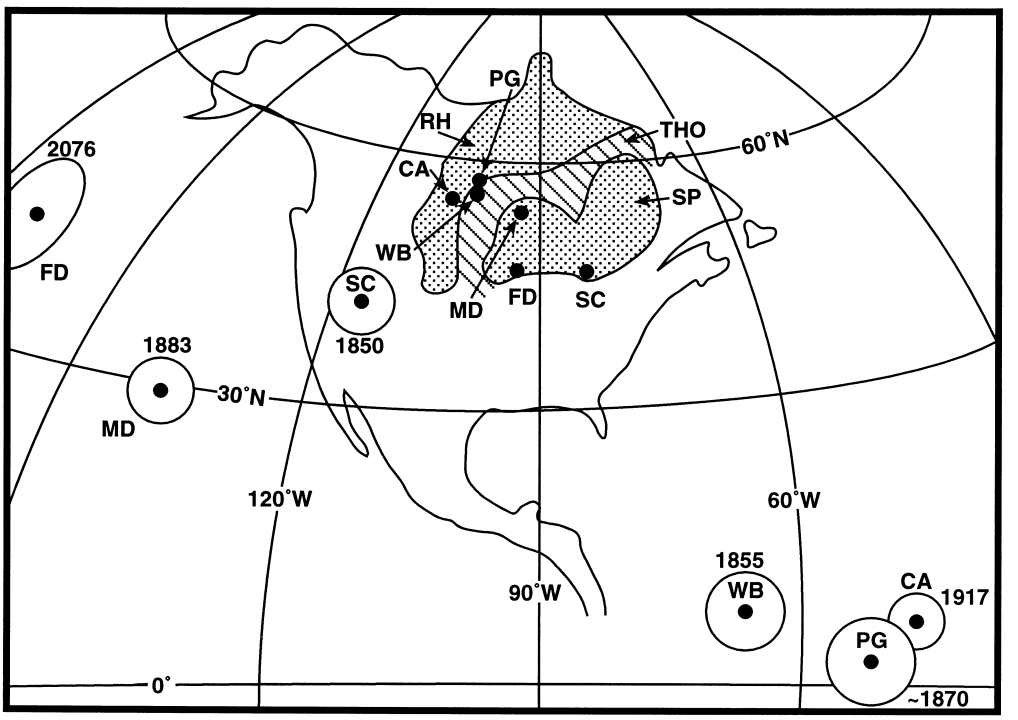


Fig. 10. A comparison of 1850–2080 Ma paleomagnetic pole positions from the Superior (*SP*) and Rae/Hearne (*RH*) provinces, together with their corresponding sampling locations. Numbers beside poles are ages in Ma. Poles: *PG* = Peter Lake gabbro and assigned age (Symons, 1994); *WB* = Wathaman batholith (Symons, 1991); *CA* = Clearwater anorthosite (this study); *FD* = Fort Frances dyke swarm (Halls, 1986; Buchan et al., 1996); *SC* = Sudbury Complex (mean of norite and granophyre-1 intrusions, Morris, 1984); *MD* = **B** remanence dykes of the Molson swarm (Zhai et al., 1994). Note that the age of the **B** pole of Zhai et al. (1994) is now given as 1883 Ma. In Zhai et al. (1994) a **B** dyke is exposed at site 89 along the bank of the Nelson River which follows a channel formed by the erosion of the giant Nelson River dyke. A dyke in Cross Lake, about 100 km to the south, yields a U–Pb date of 1883 Ma and is thought to be an extension of the Nelson River dyke (Heaman et al., 1986). Only older dykes are known in the Nelson River area; a dyke in neighbouring Cauchon Lake, one of several ENE–WSW-trending diabase dykes carrying a **C** magnetization (Zhai et al., 1994), gives a U–Pb age of 2092 Ma (Halls and Heaman, 1997). Therefore, the dyke at site 89 is likely to be either an apophysis of the Nelson River dyke or has been magnetically reset by it, thereby dating **B** at 1883 Ma.

300ºC) of about 2.5ºC/Ma. Alternatively, the cooling was more rapid, and the 1840 Ma biotite closure age is a result of resetting of the biotite (but not the hornblende) during a later event that is linked to the emplacement of the 1855 Ma Wathaman batholith.

It is clear that the **A** magnetization does not date from the time of original crystallization of the Clearwater anorthosite because the **A** direction remains constant despite deformation of the layering at site 15 which is particularly revealed by the AMS data (Fig. 8), and by the total lack of correlation between **A** and the attitude of foliation either as mapped or as given by the AMS data. Since it is likely that the anorthosite was originally formed at lower crust or even upper mantle depths, the observed ductile deformation likely occurred when the anorthosite was still at elevated temperatures corresponding to these large depths. The probability that the magnetization is partly carried by sub-microscopic magnetite in the feldspar suggests that it formed after slow cooling, perhaps as a thermo-chemical remanence when or after the magnetite exsolved. Therefore, **A** could date from the time following the original uplift of the anorthosite when the granulite-grade rocks cooled through temperatures of about 500–600ºC. Given that this temperature is similar to that of argon closure in hornblende, and assuming that the remanence is substantially a TRM (we have no evidence

otherwise) our argon–argon age of ~1.92 Ga is interpreted as the best estimate of the time of acquisition of **A**, with the ~1.84 Ga argon–argon closure age for biotite providing a lower limit.

The **A** pole position lies within ~5° of the pole (Plat. = 3°N, Plong. = 52°W, dp = 7°, dm = 12°) for the Peter Lake gabbro (Symons, 1994), an intrusion in the Archean Peter Lake Domain (Fig. 1). The 1920 Ma age for **A**, and the virtually identical positions of the Clearwater anorthosite and Peter Lake gabbro poles, support the Symons (1994) contention that the Peter Lake gabbro pole is related to deformation in the Peter Lake Domain that preceded intrusion of the Wathaman batholith.

As shown in Fig. 10, the Clearwater anorthosite pole, together with those from the Peter Lake gabbro and Wathaman batholith (Symons, 1991) form a cluster of poles (dated between 1855 and 1917 Ma) with a significantly different position from a cluster of paleomagnetically and geochronologically well-constrained poles (Sudbury Complex, Molson dykes and Fort Frances dykes) from the Superior Province on the opposite side of the THO which have a similar age range (1850–2080 Ma). Our new data strengthen the case for the existence of the Manikewan ocean because the Clearwater anorthosite pole comes from an uplifted granulite terrane of the Archean Rae Province that is clearly outside the THO and has a Ar–Ar hornblende age that probably dates the remanence. One unknown aspect of **A** is whether or not it has suffered tectonic tilting. The similarity in the **A** direction throughout the sampled region in Fig. 3, the lack of brittle deformation and the close correspondence of the pole with that from the Peter Lake Domain all suggest that later deformation has not been significant.

Our original objective, to test if the Superior and Rae/Hearne provinces were ever joined, was not attained because the Clearwater anorthosite yielded a much younger remanence than 2.1 Ga, the estimated time of rifting that initiated the Manikewan ocean.

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Appendix A. Tables of stepwise gas release data for argon–argon analyses — errors are ±2σ

CL11-11 Hornblende (*J* value: 0.027873 ± 0.000028)

Power (W)	Volume $^{39}Ar_K$ ($\times 10^{-12}$ cm^3 NTP)	Volume $^{37}Ar_{Ca}$ ($\times 10^{-12}$ cm^3 NTP)	Atmos (%)	$^{40}Ar^*/^{39}Ar_K$	Date (Ma)
3.00	252.99 ± 1.84	1249.78 ± 116.2	1.32	86.199 ± 0.851	2209 ± 13
3.75	150.85 ± 0.94	581.69 ± 72.18	0.74	67.614 ± 0.595	1911 ± 10
4.25	64.53 ± 0.54	223.70 ± 31.77	0.65	68.552 ± 0.697	1928 ± 12
5.00	35.76 ± 0.13	107.41 ± 22.91	0.52	68.014 ± 0.668	1918 ± 12
8.00	153.21 ± 1.00	619.59 ± 77.50	0.90	67.992 ± 0.616	1918 ± 11

CL15-3 Biotite (J value: 0.027870 ± 0.000054)

Power (W)	Volume $^{39}Ar_K$ ($\times 10^{-13}$ cm^3 NTP)	Volume $^{37}Ar_{Ca}$ ($\times 10^{-13}$ cm^3 NTP)	Atmos (%)	$^{40}Ar^*/^{39}Ar_K$	Date (Ma)
0.75	270.12 ± 2.60	19.72 ± 162.70	4.34	50.633 ± 0.738	1588 ± 15
1.25	794.37 ± 2.87	551.30 ± 312.62	0.98	63.389 ± 0.681	1836 ± 12
1.75	902.50 ± 6.49	611.16 ± 356.93	0.66	64.190 ± 0.645	1850 ± 12
2.25	851.26 ± 6.42	626.04 ± 354.16	0.71	63.702 ± 0.723	1842 ± 13
2.75	572.65 ± 4.19	280.65 ± 247.12	0.86	63.366 ± 0.703	1836 ± 13
3.50	734.06 ± 5.26	499.84 ± 262.00	0.63	63.112 ± 0.639	1831 ± 12
4.25	584.27 ± 4.41	351.78 ± 244.22	0.69	63.629 ± 0.698	1840 ± 13
7.00	1177.80 ± 9.00	176.48 ± 56.47	0.73	63.702 ± 0.684	1842 ± 12

FL3-11 Biotite (J value: 0.027877 ± 0.000058)

Power (W)	Volume $^{39}Ar_K$ ($\times 10^{-13}$ cm^3 NTP)	Volume $^{37}Ar_{Ca}$ ($\times 10^{-13}$ cm^3 NTP)	Atmos (%)	$^{40}Ar^*/^{39}Ar_K$	Date (Ma)
0.75	46.17 ± 0.92	−3.35 ± 153.71	13.11	43.562 ± 1.880	1434 ± 43
1.25	226.39 ± 2.36	−19.25 ± 162.14	1.63	61.346 ± 0.889	1799 ± 17
1.75	474.67 ± 4.01	165.07 ± 192.53	0.79	64.008 ± 0.856	1847 ± 15
2.25	598.75 ± 4.76	265.50 ± 246.71	0.83	63.420 ± 0.779	1837 ± 14
2.75	490.25 ± 3.79	245.30 ± 206.20	0.68	63.708 ± 0.765	1842 ± 14
3.50	468.11 ± 3.54	206.13 ± 210.07	0.59	63.722 ± 0.737	1842 ± 13
4.25	440.01 ± 3.95	193.69 ± 198.11	0.59	63.907 ± 0.817	1846 ± 15
7.00	1689.58 ± 13.71	1399.88 ± 672.10	0.88	63.188 ± 0.744	1833 ± 14

FL3-3-2 Biotite (J value: 0.027878 ± 0.000060)

Power (W)	Volume $^{39}Ar_K$ ($\times 10^{-13}$ cm^3 NTP)	Volume $^{37}Ar_{Ca}$ ($\times 10^{-13}$ cm^3 NTP)	Atmos (%)	$^{40}Ar^*/^{39}Ar_K$	Date (Ma)
1.20	184.69 ± 2.08	43.26 ± 154.18	5.50	51.017 ± 0.845	1596 ± 18
2.20	760.34 ± 6.60	640.71 ± 293.46	0.77	62.012 ± 0.740	1811 ± 14
7.00	2715.81 ± 26.35	2758.89 ± 731.90	0.73	61.562 ± 0.803	1803 ± 15

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